

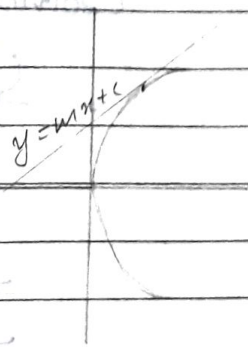
2 tangents are possible from one point.

Tangents and normals

Tangents in terms of the slope m

Q. Find the condition that the line $y = mx + c$ may touch:

- (i) parabola
- (ii) ellipse
- (iii) hyperbola.



- (i) Given $y = mx + c$ — (1)
- and parabola, $y^2 = 4ax$ — (2)

Solving (1) and (2) we can find the point of intersection.

$$m^2x^2 + c^2 + 2mxc = 4ax$$

$$m^2x^2 + 2mxc - 4ax + c^2 = 0$$

$$m^2x^2 + x(2mc - 4a) + c^2 = 0 \quad \text{--- (3) } (b^2 - 4ac = 0)$$

which is a quadratic equation in x . Then eqn (1) will touch the parabola if the roots of the eqn (3) are equal.

$\therefore b^2 - 4ac$ should be equal to zero.

$$\Rightarrow (2mc - 4a)^2 - 4m^2x^2c^2 = 0$$

$$4m^2c^2 - 16mac - 4m^2x^2c^2 + 16a^2 = 0$$

$$16a^2 - 16mac = 0$$

$$16a(a - mc) = 0$$

$$a - mc = 0$$

$c = a/m$ which is the req. condition.

$\therefore$ The line $y = mx + a/m$ is a tangent to the parabola $y^2 = 4ax$.

(ii) Given $y = mx + c$ — (1)

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ — (2)

Solving (1) and (2) we can find the point of intersection

$\Rightarrow \frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$

$\Rightarrow b^2 x^2 + a^2 (mx+c)^2 = a^2 b^2$

$\Rightarrow b^2 x^2 + m^2 a^2 x^2 + 2mxc a^2 + a^2 c^2 = a^2 b^2$

$\Rightarrow x^2 (b^2 + m^2 a^2) + x (2m a^2 c) + a^2 (c^2 - b^2) = 0$

eqn (1) will touch eqn (2) when eqn (3) has equal roots

$\Rightarrow b^2 - 4ac = 0$

$\Rightarrow (2m a^2 c)^2 - 4(b^2 + m^2 a^2)(a^2 c^2 - b^2) = 0$

$\Rightarrow 4m^2 c^2 a^4 - 4(b^2 + m^2 a^2)(a^2 c^2 - b^2) = 0$

$\Rightarrow 4m^2 c^2 a^4 - 4a^2 c^2 b^2 + 4a^2 b^4 - 4m^2 a^4 c^2 + 4m^2 a^2 b^2 = 0$

$\Rightarrow 4m^2 a^4 b^2 - 4a^2 c^2 b^2 + 4a^2 b^4 = 0$

$\Rightarrow 4a^2 b^2 (m^2 a^2 - c^2 + b^2) = 0$

$\Rightarrow m^2 a^2 - c^2 + b^2 = 0$

$\Rightarrow c^2 = m^2 a^2 + b^2$

$\Rightarrow c = \pm \sqrt{m^2 a^2 + b^2}$

$\therefore$ the line $y = mx \pm \sqrt{m^2 a^2 + b^2}$ is a tangent to the ellipse

(iii) Given $y = mx + c$ — (1)

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ — (2)

Solving (1) and (2) we can find the point of intersection

$$\frac{x^2}{a^2} - \frac{(mx+c)^2}{b^2} = 1$$

$$b^2x^2 - a^2(mx+c)^2 = a^2b^2$$

$$b^2x^2 - a^2m^2x^2 - 2mxca^2 - a^2c^2 = a^2b^2$$

$$x^2(b^2 - a^2m^2) + x(-2mca^2) - a^2c^2 - a^2b^2 = 0 \quad \text{--- (3)}$$

The line equ(1) will touch equ(2) when equ(3) has equal roots.

$$\text{ie } b^2 - 4ac = 0$$

$$\Rightarrow 4m^2c^2a^4 + 4(a^2c^2 + a^2b^2)(b^2 - a^2m^2) = 0$$

$$\Rightarrow 4m^2c^2a^4 + 4a^2c^2 + 4a^2b^2(b^2 - a^2m^2) = 0$$

$$\Rightarrow 4m^2c^2a^4 + 4a^2b^2c^2 - 4a^4m^2c^2 + 4a^2b^4 - 4a^4m^2b^2 = 0$$

$$\Rightarrow 4a^2b^2c^2 + 4a^2b^4 - 4a^4m^2b^2 = 0$$

$$\Rightarrow 4a^2b^2(c^2 + b^2 - a^2m^2) = 0$$

$$\Rightarrow c^2 = a^2m^2 - b^2$$

$$\Rightarrow c = \pm \sqrt{a^2m^2 - b^2}$$

the line $y = mx \pm \sqrt{a^2m^2 - b^2}$ is tangent to the hyperbola $\forall m$.

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Q. S. + Two tangents can be drawn from any point to
(a) parabola (b) ellipse (c) hyperbola

(a) We know that $y = mx + \frac{a}{m}$ is a tangent to the parabola $y^2 = 4ax$, $\forall m$. If this pass through any point (x_1, y_1) we have

$$\text{then } y_1 = mx_1 + \frac{a}{m}$$

$$my_1 = m^2x_1 + a$$

$$m^2x_1 - my_1 + a = 0 \quad \text{--- (1)}$$

The eqn (1) is a quadratic in m and hence gives two values for m which in turn gives a two tangents from the point (x_1, y_1) .

Hence
Note: The two tangents from (x_1, y_1) will be real, coincident or even imaginary according to the roots of eqn (1) are real/imaginary i.e., according as the point (x_1, y_1) is outside/inside the parabola.

(b) We know that $y = mx \pm \sqrt{m^2 a^2 + b^2}$ is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $\forall m$. If this pass through any

point (x_1, y_1) ,

$$\text{then } y_1 = mx_1 \pm \sqrt{m^2 a^2 + b^2}$$

$$(y_1 - mx_1)^2 = m^2 a^2 + b^2$$

$$m^2(x_1^2 - a^2) - 2mx_1y_1 + y_1^2 - b^2 = 0 \quad \text{--- (2)}$$

eqn (2) is a quadratic in m , $\Rightarrow$ gives two values for m .
 $\therefore$ two tangents can be drawn from one point (x_1, y_1)

(c) We know that $y = mx \pm \sqrt{m^2 a^2 - b^2}$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $\forall m$. If this pass through any point (x_1, y_1)

$$\text{then } y_1 = mx_1 \pm \sqrt{m^2 a^2 - b^2}$$

$$(y_1 - mx_1)^2 = m^2 a^2 - b^2$$

$$m^2(x_1^2 - a^2) - 2mx_1y_1 + y_1^2 + b^2 = 0 \quad \text{--- (3)}$$

eqn (3) is a quadratic in m $\Rightarrow$ it gives two values for m .
 $\therefore$ two tangents can be drawn from one point (x_1, y_1)

Date ____ / ____ / ____

Saathi

Q. Find the condition for ellipse $lx + my + n = 0$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$lx + my + n = 0$$

$$y = -\frac{lx + n}{m}$$

$$= -\frac{l}{m}x - \frac{n}{m}$$

$$y = mx + c$$

$$\therefore M = -l/m, \quad c = -n/m$$

The condition for ellipse is

$$c^2 = m^2 a^2 + b^2$$

$$\frac{n^2}{m^2} = \frac{l^2}{m^2} a^2 + b^2$$

$$\underline{\underline{n^2 = l^2 a^2 + m^2 b^2}} \quad \text{is the new condition}$$